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Short Communication

Shotgun metagenomic analysis reveals the emergence of plasmid-encoded *mcr-5.1* gene in hospital wastewater in Bangladesh

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ABSTRACT

Colistin is considered the last line therapy for treating multidrug-resistant (MDR) bacterial infections in humans. Therefore, the spread of colistin resistance poses a serious threat to human, and environmental health. Though Bangladesh is known as a hotspot of AMR, limited studies have been carried out regarding the status of colistin resistance. Information on the emerging bacterial resistance is inevitable for protecting public health. Nowadays, wastewater analysis has been prioritized for metagenomics-enabled AMR surveillance. Our study on the metagenomic analysis of untreated hospital effluents first detected the colistin resistance-conferring *mcr-5.1* gene in the hospital environment of Bangladesh. Phylogenetic tree and *in silico* AMR analysis confirmed the detection of this *mcr-5* variant, which is located in a plasmid contig. The plasmid was untypeable and belonged to the bacteria from the Enterobacteriaceae family. The *mcr-5.1* operon was embedded in a Tn3 transposon, suggesting the mobility of the gene. *Tnshfr1* transposon, chromate resistance protein ChrB, DNA invertase hin, and two MFS-type proteins were present in the genetic environment of *mcr-5.1*. Our findings provide evidence of the occurrence of *mcr-5.1* in a hospital environment in Bangladesh, which calls for immediate attention and effective measures to contain the dissemination of colistin resistance in the environment.

Introduction

Colistin, a multicomponent cationic polypeptide antibiotic, disrupts bacterial cell-membrane by binding to the anionic lipopolysaccharide (LPS) molecules in the gram-negative bacteria's outer membrane through electrostatic interactions and, thus, exerts its antimicrobial effect [1]. It is exclusively used in veterinary not only to control infections by Enterobacteriaceae but also to act as a poultry growth promoter [2]. In recent years, it has gained considerable attention as the drug of last resort and is increasingly used for controlling human infections caused by carbapenemase-producing Enterobacteriaceae, *Pseudomonas aeruginosa, Acinetobacter baumanii* and others [3]. Colistin resistance can be occurred by chromosomal mutations and by the acquisition of plasmid-carrying determinants. Plasmid-mediated mobile colistin resistance (*mcr*) genes are strongly linked to phenotypic colistin resistance [4]. Ten *mcr* genes and multiple variants have been found worldwide from different sources, including humans, animals and environment, since the plasmid-mediated colistin resistance mechanism was first identified in December 2015 [5–7]. Epidemiological studies on the distribution of the *mcr* gene showed that *mcr-5* is one of the most prevalent variants among the *mcr* family [5]. It was first detected in *Salmonella* Paratyphi B var. Java dTa+ isolated from Germany and till now, 4 variants of *mcr-5* have been detected [8]. MCR-5 has only 34%–36% amino acid sequence similarity with MCR-1, MCR-2, MCR-3, and MCR-4, making it unique from the other four proteins [9].

Like other developing countries, antimicrobial resistance has emerged as a severe concern in Bangladesh. Resistant bacteria from livestock can transfer to humans via the food chain, posing a threat to human health and food safety. Previous studies have reported the detection of the variants of *mcr* (*mcr-1* and *mcr-8*) in clinical and environmental pathogens in Bangladesh [4,10]. Such occurrence of AMR pathogens might be attributed to unrestricted use of antibiotics. Prior culture-based studies showed that hospital wastewater is one of the major drivers in the emergence and dissemination of AMR in Bangladesh [11]. Nowadays, wastewater

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testing is being growingly recognized for tracking the transmission of AMR genes and pathogens in the environment and also for disease prediction. Besides, shotgun metagenomics has emerged as a powerful tool for characterising AMR in wastewater microbial communities over conventional culture-based methods. Therefore, we conducted a shotgun metagenomic analysis of untreated hospital effluent to depict the comprehensive scenario of AMR prevailing in the hospital environment. Based on our findings, here, we present the first report on the detection of the *mcr-5.1* gene in hospital wastewater in Bangladesh. This study acts as an early warning indicator on colistin resistance that will help the clinicians, experts and the government to set up a multisectoral plan and strategic framework to limit the development and transmission of AMR.

Materials and methods

Sample collection

Hospital wastewater samples were collected from 7 divisional tertiary-level hospitals all over Bangladesh between July 2022 and November 2022 (Supplementary Table 1). 1.5 L of untreated sewage water was collected in sterile glass bottles from three different points of the central sewage system that receives effluents from different hospital sites. Samples were transported to the laboratory in an ice box with ice and processed within 12 hrs of collection. 50 ml of the sample was centrifuged at 8000 rpm for 10 min, and then, the resulting pellet was stored at −20 °C before DNA extraction.

DNA extraction and metagenomic sequencing

The DNA was extracted using DNeasy PowerSoil Pro Kit (Qiagen) according to manufacturer's instructions. The extracted DNA was checked for quality and concentration using NanoDrop spectrophotometer. Following library preparation using Nextera XT DNA Library Preparation Kit, shotgun metagenomic sequencing was carried out on an Illumina NovaSeq 6000 sequencer (paired-end, 2 × 150 bp) at EzBiome, USA.

Pre-processing and assembly of raw data

The quality of both raw and trimmed sequences was evaluated using FastQC (https://www.bioinformatics.babraham.ac.uk/projects/fastqc/). fastp [12] was used to remove the adapter and low-quality reads. The trimmed reads were assembled using MEGAHIT with default parameters [13]. In addition, the raw fastq files were implemented in metaplasmidSPAdes [14] in order to detect and assemble plasmids in the metagenome. Plasflow [15] and PlasClass [16] were used to confirm the identity of the sequence as a plasmid.

Taxonomic classification and metagenome annotation

Taxonomic classification of contigs was determined using Kraken2 against the NCBI nucleotide non-redundant (nt/nr) database [17]. Prokka V1.14.5 integrated in KBase (https://www.kbase.us/) was utilised to annotate the rest of the contigs [18]. The host taxonomy of plasmid contig was predicted using HOTSPOT that exploits transformer, a state-of-the-art language model for accurate prediction of plasmid's host [19].

Detection of mcr-5.1 *and its origin*

We used ABRicate (https://github.com/tseemann/abricate) to identify antimicrobial resistance genes by mass screening of contigs using CARD [20], and ResFinder [21] (minimum DNA identity: 80%, minimum DNA coverage: 80%). The *mcr-5.1* carrying plasmid contig was BLAST searched to define the genetic environment. The insertion sequence in the plasmid was identified by ISFinder [22]. The whole plasmid contig was aligned with closely related sequences using MEGA 11 to construct a phylogenetic tree (Bootstrap replicates = 1000). Finally, the genomic environment surrounding *mcr-5.1* of our plasmid sequence was compared with other bacteria using the Easyfig tool.

Results

In order to characterize the diversity in resistome prevailing in hospital environments, hospital wastewater samples were collected from seven divisional hospitals in Bangladesh. According to our analysis, the *mcr-5.1* was detected with a relative abundance of 0.42% in the metagenome extracted from the Sher-e-Bangla Medical College Hospital (S.B.M.C.H.) (Sample code-W1), Barisal, Bangladesh (22°41′11.0″N, 90°21′40.7″E).

Assembly of quality filtered raw sequences using MEGAHIT and subsequent screening of assembled contigs using ABRicate against different databases identified the presence of *mcr-5.1* gene in the contig >k141_2312 (5345 bp). The region ranging from 1456 to 3099 bp (coverage length 1–1644/1644) of the contig was predicted to encode phosphoethanolamine—lipid A transferase MCR-5.1 that showed 100% coverage and 100% identity with Accession ID KY807921 (ResFinder), NG_055658.1 (NCBI), MG241339:1–1644 (ARG-ANNOT) and KY807920.1:11,539–9895 (CARD). In addition, extraction and assembly of plasmid by metaplasmidSPAdes showed that the *mcr-5.1* gene was located in contig NODE_230 (7486 bp). NODE_230 was further identified as an untypeable plasmid sequence belonging to proteobacteria with a score of 0.9169548. Fig. 1 presents the genetic environment of *mcr-5.1* in both contig and plasmid.

The BLAST search of the plasmid sequence showed 100% identity with *Escherichia coli* strain ENV103 plasmid pSGMCR103 (Accession: MK731977.1) at 98% query coverage. The total length of the alignment was 7338 bp, in which, the contig's nucleotides from 73 to 7410 bp aligned with 6384 to 13,721 bp of *E. coli* plasmid. No gaps were found in the alignment. Such identity in the *mcr-5.1* gene and surrounding regions further confirmed the presence of this gene on a plasmid that closely resembles those found in *E. coli* (Fig. 2). In addition, 37 chromosomal and plasmid sequences of *Aeromonas veronii, Salmonella* Paratyphi B, *Escherichia fergusonii, Cupriavidus gilardii, Enterobacter mori, Aeromonas hydrophila* etc. exhibited >90% identity and coverage with the query sequence.

Phylogenetic analysis clustered all *mcr* variants into two distinctive clades, and our predicted gene shared the same clade with reference *mcr-5.1* (NCBI Reference Sequence: NG_055658.1) (Fig. 3A). In addition, the constructed phylogenetic tree showed that *E coli* plasmid pSGMCR103 and *Cupriavidus gilardii* were the closest neighbours of the retrieved plasmid (Fig. 3B).

Taxonomic classification of the plasmid sequence by KRAKEN 2 assigned it to the Enterobacteriaceae family. On the other hand, the complete host lineage output by HOTSPOT for the plasmid contig was Phylum: Pseudomonadota, Class: *Gammaproteobacteria*, Order: *Enterobacterales*, Family: Enterobacteriaceae, Genus: *Salmonella*, Species: *Salmonella enterica*. (from phylum to species). However, the result of the accurate mode of HOTSPOT was Phylum: Pseudomonadota, Class: *Gammaproteobacteria*, Order: *Enterobacterales*.

The analyses of the genetic environment of the *mcr-5.1* that the gene was located within chromate B (*chrB*) and a multidrug resistance protein MdtH, a major facilitator superfamily (MFS) type transporter. PROKKA annotation identified a TnShfr1 (Tn3 family transposase) at the upstream of *mcr-5.1* which was further confirmed by ISFinder. The comparison of the *mcr-5.1* gene and its

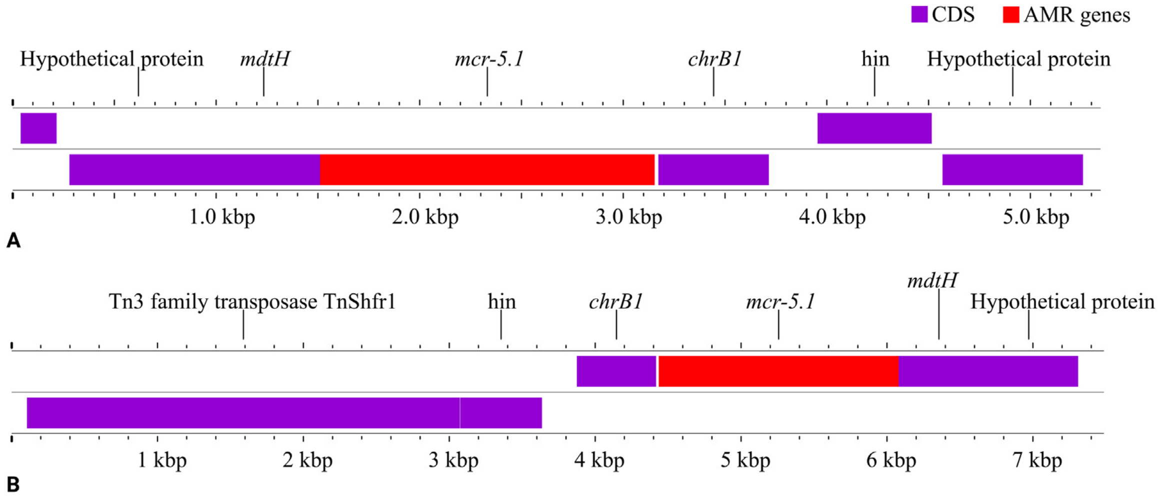

Fig. 1. Annotation of *mcr-5.1* carrying contig (A) and plasmid (B).

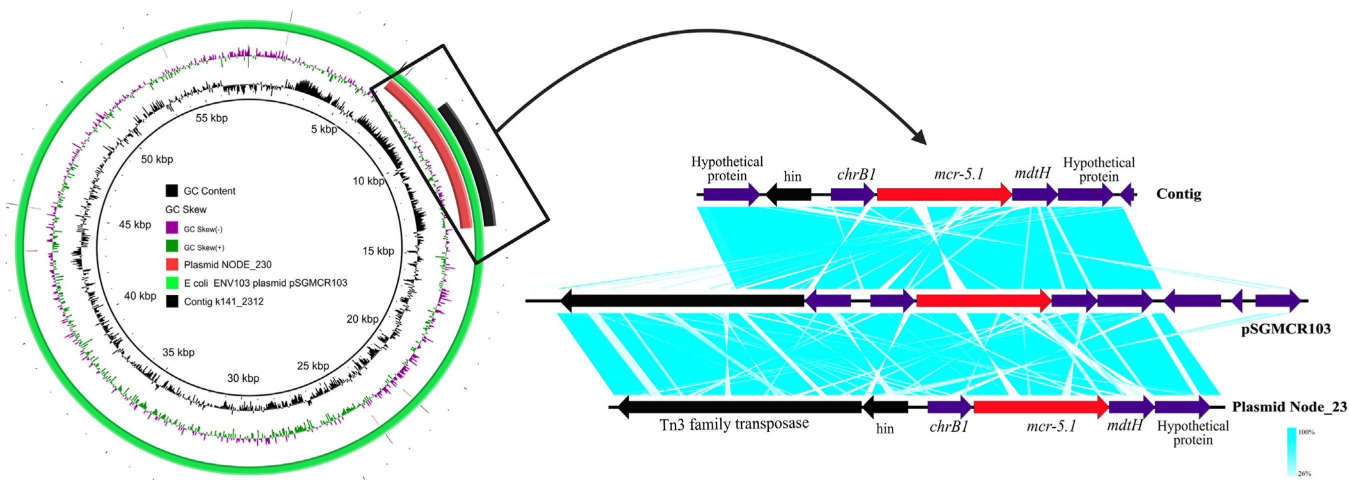

Fig. 2. Comparison of the contig and plasmid sequence with *E. coli* ENV103 pSGMCR103 (The red, black and purple arrows represent *mcr-5.1* gene, transposable elements and hypothetical proteins, respectively in the linear comparison of sequences).

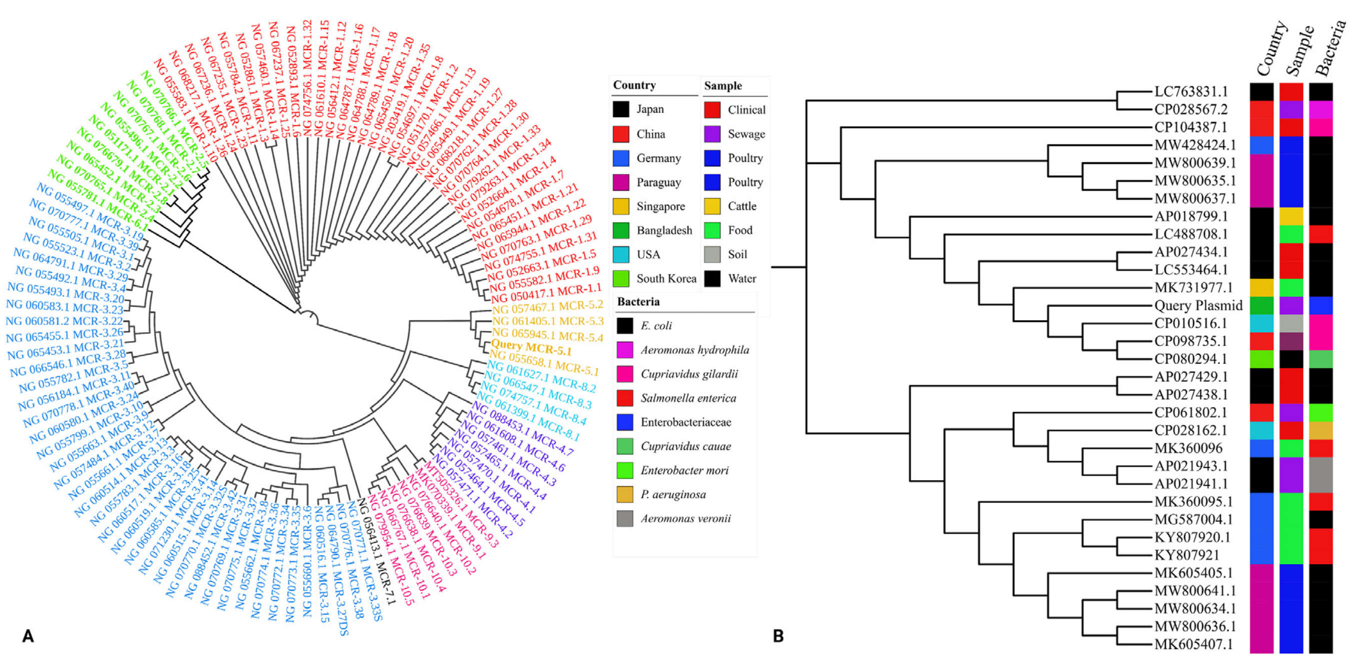

Fig. 3. Phylogenetic tree constructed using (A) all *mcr* variants and (B) retrieved plasmid sequence with similar sequences.

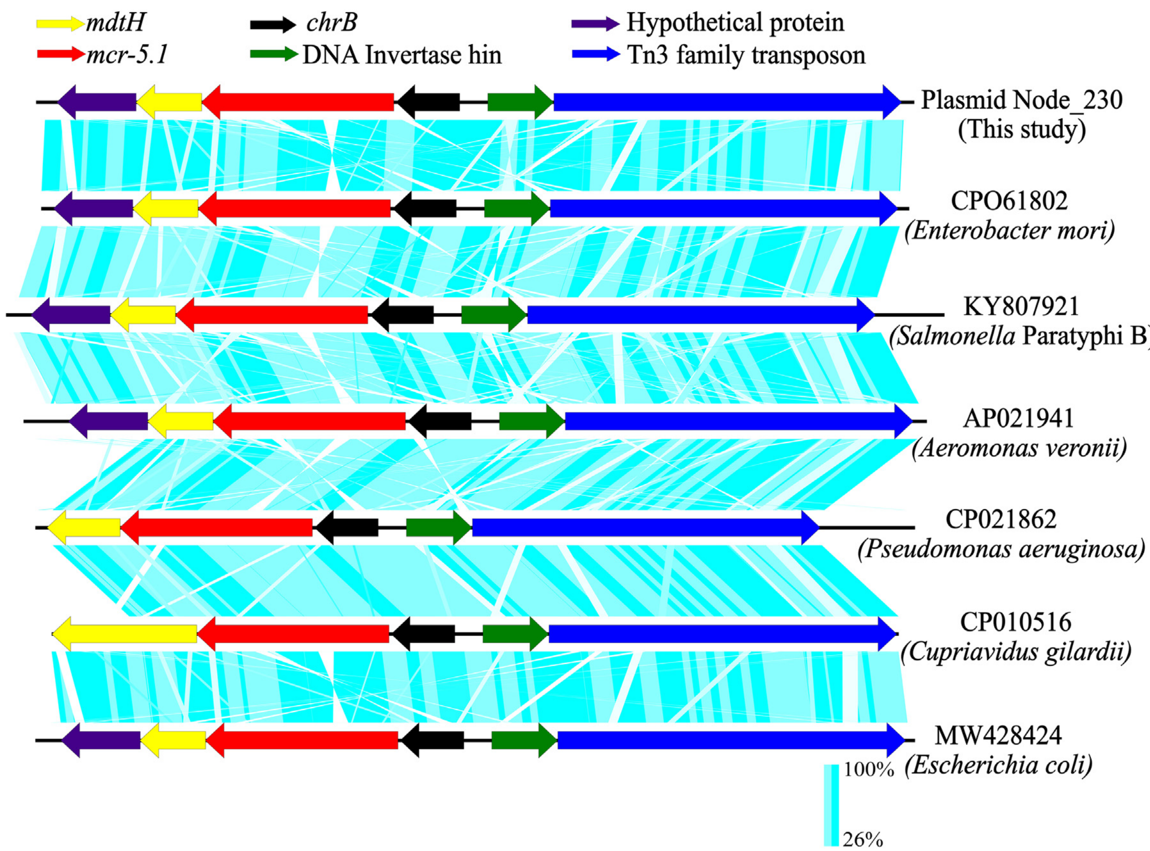

Fig. 4. Comparison of the genetic environment of *mcr-5.1* in chromosome and plasmid sequences from different bacteria.

surrounding environment in plasmids and chromosomes of different bacteria also showed similar genomic architecture across *Enterobacter mori, Salmonella* Paratyphi B, *Aeromonas veronii, Pseudomonas aeruginosa, Cupriavidus gilardii* and *E. coli* (Fig. 4). The consistent presence of such gene arrangements across diverse bacterial species suggests a common mechanism of horizontal gene transfer.

Discussion

A 1644 bp long *mcr-5.1* was identified in the wastewater metagenome collected from Sher-E-Bangla Medical College, Barisal, Bangladesh. According to the Health Bulletin 2020 published by the Ministry of Health and Family Welfare, Government of the People's Republic of Bangladesh, the number of outpatients, emergency and inpatients who attended this 1500 bedded tertiary level hospital in 2020 were 251,072, 110,827 and 101,638, respectively [23]. Patients from surrounding districts can attend this hospital facility for diagnosis and treatment. Therefore, such occurrence of *mcr-5* gene in the hospital environment might make healthcare professionals, patients and visitors vulnerable to colistin-resistant bacterial infections.

Though the chromosomal origin of *mcr-5.1* was reported, it is predominantly located in plasmid [24]. Similarly, *the mcr-5.1* carrying sequence in our study was also identified as a plasmid. As plasmids can be transferred between bacteria, the identification of plasmid host ranges contributes to understanding its transmission in bacterial communities as well as its role in bacterial evolution and adaptability [19]. According to taxonomic assignment and *in silico* host prediction tool, the *mcr-5.1* harbouring plasmid might be hosted by bacterial species of the Enterobacteriaceae family. In addition, the phylogenetic analysis suggested that this plasmid was clonally related to *E coli* plasmid pSGMCR103 isolated from RTE food in Singapore. pSGMCR103 is an IncX1 type narrow host range plasmid that is commonly found in Enterobacteriaceae [25]. A recent study from our neighbouring country, India, also reported the occurrence of plasmid-encoded *mcr-5.1*, belonging to *E. coli* pSGMCR103 plasmid in hospital wastewater [9]. Along with our study, some other recent articles also found this gene in the hospital environment [9,24]. A comparison of this evidence suggested the plasmid-mediated environmental transmission of *mcr-5.1* by horizontal gene transfer (HGT). Therefore, active surveillance is crucial to keep track of the emergence of new variants and the route of transmission.

The surrounding environment of *mcr-5.1* consisted of Tn3 family transposase TnShfr1, ChrB protein, DNA invertase, and two uncharacterized MFS transporters. The arrangement of these components was similar to a study that first described the location of *mcr-5.1* on a 7337 bp Tn3-type transposon flanked by 38 bp IRs in *Salmonella* Paratyphi B isolate 13-SA01718 plasmid pSE13-SA01718 [8]. A recent study from Japan also reported the TnShfr1 transposon in the *mcr-5.1* carrying pN-ES-6–2 plasmid in *E. coli* [26]. Borowiak et al. suggested that colistin resistance mediating gene *mcr-5.1* might be transferred from bacterial chromosomes to mobile elements like plasmid [8]. Our analysis of the comparison of the *mcr-5.1* genomic environment indicated the conservation of *chrB-mcr5.1*-MFS fragments in the plasmid and chromosome of other bacteria. This finding was in agreement with Xu et al., who also reported the high conservancy of this fragment and suggested it as the core transmission unit of *mcr-5* containing plasmid [27]. Moreover, this fragment was also implicated in the high stability of *mcr-5* operon as described by Ma et al. [28]. Overall, it can be said that the genetic organization of *mcr-5.1* carrying operon might have an important role in the stability of the plasmid and transmission of this gene.

In Bangladesh, the direct release of untreated hospital wastewater containing AMR genes and pathogens into municipal sewage systems and water bodies causes pollution of the surrounding environment. A recent study reported the failure of waste treatment processes in removing colistin resistance genes. Therefore, leakage of wastewater into groundwater and use of contaminated water in irrigation may facilitate the entry of *mcr-5.1* into agriculture and food production. Considering the clinical outcomes of colistin resistance, its impact in Bangladesh will be deleterious because of high population density, poor waste management, self-prescription of antibiotics, faster spread of plasmid mediated resistance gene and others. Integration of high throughput sequencing with wastewater monitoring and designing novel waste treatment strategies are essential to tackle the ever-increasing AMR issues.

Conclusion

To our knowledge, this is the first report on the emergence of the plasmid-encoded *mcr-5.1* in a hospital environment in Bangladesh. Though our analysis confirmed the detection of *mcr-5.1*, the lack of multiple sampling in different seasons remained one of the limitations of the study which might better characterize seasonal variations in the abundance of this gene. However, future research on the determination of the extent of transmission and the origin of the gene, identification of specific bacteria harbouring the plasmid with *mcr-5* might contribute to overcoming the limitations. Obviously, it goes without saying that resistance to colistin presents an alarming situation that needs immediate attention and actions to minimize the crisis. Evidence-based policies and regulations on antibiotic use together with effective intervention measures are of utmost importance to control the spread of resistance.

Data availability statement

The shotgun metagenomic sequences have been submitted on NCBI under the accession no PRJNA1073071 (BioProject).

Declarations

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Declaration of competing interests: The authors declare no conflict of interest.

Ethical approval: Not required.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jgar.2024.08.007.

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